

Process Engineering

Skill Pool Structure Update | May 2026

Shell R&D Technical Function (RDTF)
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This is US

An Overview of the Shell R&D Technical
Function



RDTF Core Disciplines



Chemical Sciences & Engineering

Chemistry, Kinetics & Catalysis, Reaction Engineering, and Integrated Process Development.



Subsurface Sciences & Engineering

Earth Sciences and Reservoir Science & Engineering focusing on hydrocarbon recovery.



Biological Sciences & Engineering

Ecological Science and Bioprocess/Biochemical Engineering for clean energy solutions.



Material Sciences & Engineering

Characterization, Synthesis, and Formulation Science of advanced engineering materials.



Electrical Sciences & Engineering

Energy Storage, Conversion, and Power Technology systems integration.



Computer Sciences & Engineering

Digital Enablers: Data Science, AI, Computational Science, and Mathematics.

RDTF Global Leadership



Technical Function Leaders

Selda Günsel

TF Head & Chief Technology Officer (CTO)

Ajay Mehta

Chief Engineer (CE)

- ✓ Defining strategic R&D direction globally
- ✓ Managing over \$1bln annual R&D investment
- ✓ Overseeing 3,300+ highly skilled scientists

Global Technology Hubs

 Houston Tech Center

USA

Houston Technology Center - A major core for Upstream and energy transition research.



Netherlands

Shell Technology Centre Amsterdam - Celebrating over 100 years of pioneering innovation.



India

Shell Technology Centre Bangalore - The heart of digital transformation and AI deployment.

Associated General Managers

Discipline Group	Associated General Managers
Chemical S&E	Dawid J D'Melo, HP Calis
Subsurface S&E	Marcella Dean-Elsener, Miranda Mooijer
Electrical S&E	Ilse van der Net, Ingmar Ploemen, Theo Bodewes
Materials S&E	Ed Schouten, Matthias Mundt, Sukanta Ghosh
Computer S&E	Tekin Menten, Karina Fernandez, Francesco Menapace, Seema Gopal

Global Discipline Heads (GDH)



Chemical: Sander van Bavel
"Making molecules"



Subsurface: Matthias Appel
Subsurface expertise



Biological: Damian Allen
Bioscience innovation



Materials: Rolf van Benthem
Structure & Properties



Electrical: Elizabeth Endler
Power & Devices



Computer: Detlef Hohl
Digital Enablers

Perspective of Experimentalists

The Technical Backbone

Essential contributors in all skill groups for "deep technical capabilities," integrated with scientists and engineers.

Communities of Practice:

- 👥 STCH Science Professional Group
- 👥 ETCA Monthly Testing Connect
- 👥 ECN @ STCB Network



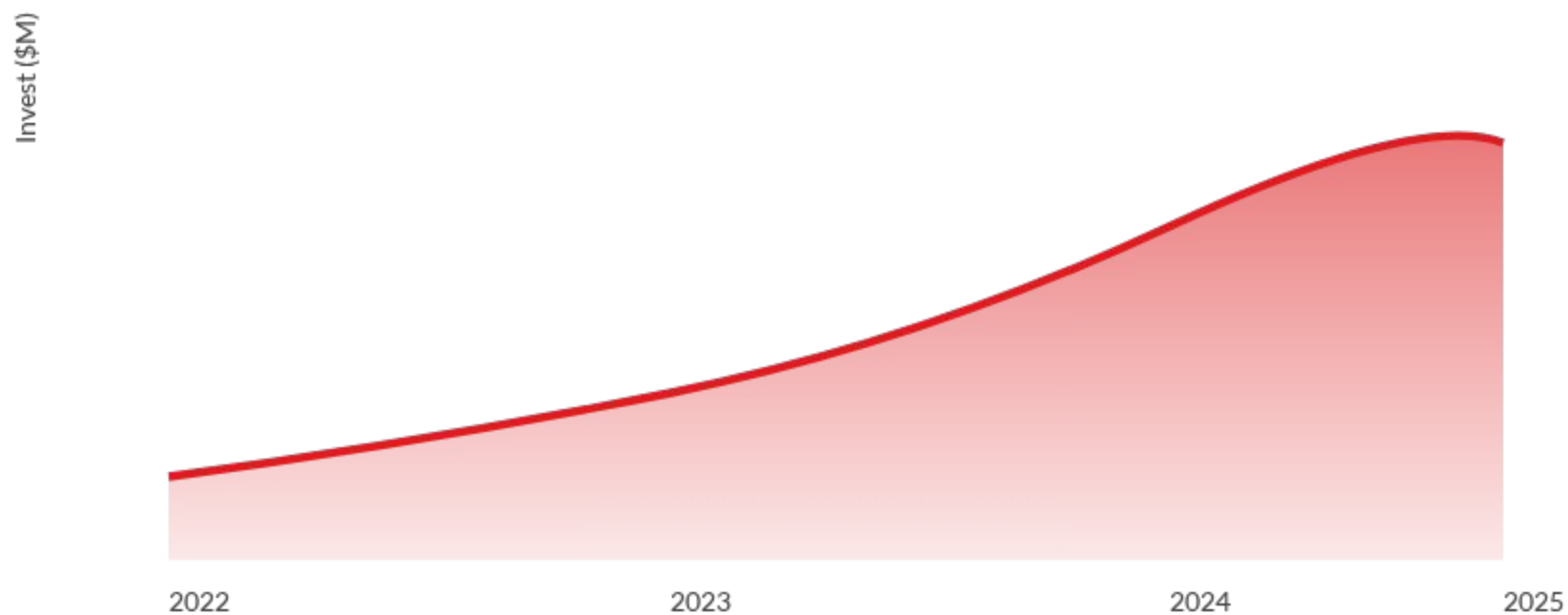
Concrete Benefits of RDTF

19 Skill Groups

Unified Value Proposition

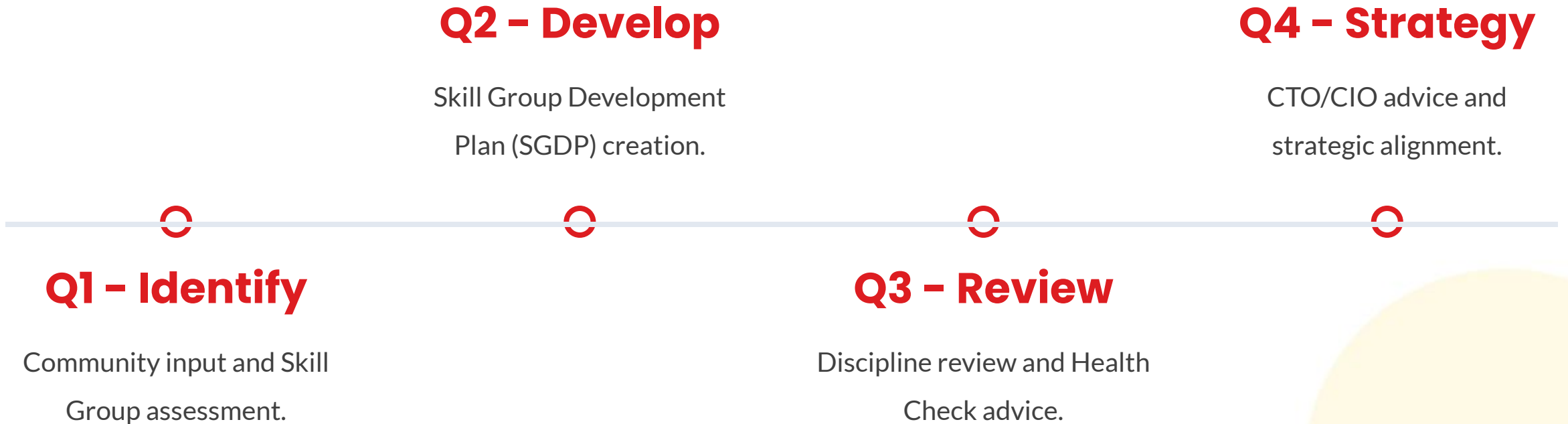
- 🛡️ **Capability:** Building "deep technical" expertise to stay future-ready.
- 🔗 **Knowledge:** Frequent, targeted sharing across disciplines.
- 📍 **Navigation:** Unified framework for skills and career growth.

Annual R&D Investment



Shell invests over **\$1 Billion** annually in R&D, with 41% focused on decarbonization projects in 2025.

RDTF Health Check: Yearly Cadence



The R&D Career Ladder



Specialist Path

Longer tenure, deep expertise in a specific scientific discipline (e.g., Chief Scientist).

Generalist Path

Broadening across R&D disciplines and management (e.g., Technology Manager/VP).

Technology Hub Hub Resources

 Collaboration and Career

Stay Connected

-  **Viva Engage:** Join the RDTF Community.
-  **SharePoint:** Access RDTF Refresh documents.
-  **Technology Webinar:** Every Wednesday.

Questions?

Process Engineering Skill Pool Structure Update

Contact: Susanne.Nielen@shell.com for Technology Webinar sign-ups.

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